Half-flat bubble exclusion beyond aspherical topology

23 September 2026. This continuation weakens the topology in the compact-core argument, and therefore in the compact-regime realization theorem. It does not alter the archived proofs. The classification of half-flat ALE spaces is a classical input; the extension below identifies the homotopical information actually needed for their exclusion.

1. The topological hypothesis

Let M be a closed connected smooth oriented four-manifold satisfying

$$\pi_1(M) \text{ is torsion-free}, \qquad \pi_2(M) = 0. \tag{H}$$

Equivalently, the second condition says $H_2(\widetilde{M};\mathbb{Z}) = 0$: the universal cover is simply connected, so this is the degree-two Hurewicz theorem. No vanishing of $\pi_3(M)$, no contractibility, and no assumption on spin are required.

Theorem 1 (compact-core exclusion under (H)). No sufficiently large compact core of a nonflat smooth Ricci-flat ALE four-manifold with one identically vanishing Weyl half embeds smoothly in a manifold satisfying (H). The assertion is independent of the orientation of the proposed embedding.

Proof. Reverse orientation on the ALE space if necessary, so that $W^+ = 0$. Let B be a large core with one connected spherical space-form boundary Y, and with the homotopy type of the ALE space. The classification of Wright, together with Suvaina's description of the Kähler ALE quotients, gives the following alternatives:

1. $\pi_1(B)$ is finite and the rational intersection form of B is negative definite, with $b_2(B) > 0$; or
2. B is a rational homology ball with

$$\pi_1(B) = H_1(B;\mathbb{Z}) = \mathbb{Z}_n, \qquad Y = L(n^2, nr-1), \quad n \geqslant 2,\ (n,r) = 1. \tag{1}$$

These are the same classification inputs used in the inherited compact-core proof. The nonflat assumption excludes the Euclidean case.

Suppose $i : B \hookrightarrow M$ is an embedding. The image of the finite group $\pi_1(B)$ in the torsion-free group $\pi_1(M)$ is trivial. Thus i lifts to an embedding into $\widetilde{M}$. The lift is injective since its composition with the covering projection is injective. This does not assert that a limiting fundamental group injects into the fundamental group of M.

In the first case a compact rational two-cycle in the interior of B has nonzero self-intersection. Its image is zero in $H_2(\widetilde{M};\mathbb{Q})$, which vanishes by (H). Intersection of compact cycles is unchanged by an embedding, up to an overall orientation sign, and is zero for a null-homologous compact cycle. This is a contradiction.

In the second case put $C = \widetilde{M} \setminus \operatorname{int} B$ and apply integral Mayer–Vietoris using collar enlargements. Since $H_1(\widetilde{M};\mathbb{Z}) = H_2(\widetilde{M};\mathbb{Z}) = 0$,

$$\mathbb{Z}_{n^2} \cong H_1(Y;\mathbb{Z}) \cong H_1(B;\mathbb{Z}) \oplus H_1(C;\mathbb{Z}) \cong \mathbb{Z}_n \oplus H_1(C;\mathbb{Z}). \tag{2}$$

The last summand has order n. The right side has exponent at most n, whereas the left side has exponent n^2. This contradiction also works for composite n. Only the first two homology groups of the universal cover were used. □

1.1 What an arbitrary remaining ALE core would have to be

There is also a classification-free part of this argument. If a connected compact oriented four-manifold B has finite fundamental group and connected rational-homology-three-sphere boundary and embeds in a manifold satisfying (H), then

$$H_2(B;\mathbb{Q}) = 0. \tag{2a}$$

Indeed the map $H_2(B;\mathbb{Q}) \to H_2(B,\partial B;\mathbb{Q})$ is an isomorphism by the boundary homology sequence. Poincaré–Lefschetz duality therefore makes the rational intersection pairing nondegenerate. The lifted embedding into $\widetilde{M}$, whose second homology vanishes, makes every such pairing zero. Non-degeneracy implies (2a). Also $H_1(B;\mathbb{Q}) = 0$ by finiteness of its fundamental group. Connected boundary and duality give $H_3(B;\mathbb{Q}) = 0$, so B is a rational homology ball.

In particular, a remaining smooth Ricci-flat ALE bubble (these have finite fundamental group and spherical-space-form ends) could not carry nonzero rational second homology. Combining this with Theorem 1 and the paired-concentration theorem, a surviving bubble would have to be a **rational homology ball with both Weyl halves nonzero and carrying positive critical mass**. This does not claim that such Ricci-flat ALE spaces exist, or exclude them by importing the classification restricted to half-flat spaces. It narrows precisely the unresolved compact-regime degeneration.

2. The enlarged class contains genuinely nonaspherical manifolds

Proposition 2. Suppose $X_1, \dots, X_N$ are closed oriented four-manifolds satisfying (H). Then

$$M = X_1 \# \dots \# X_N \# m(S^1 \times S^3) \tag{3}$$

also satisfies (H).

Proof. Van Kampen identifies $\pi_1(M)$ with the free product of the $\pi_1(X_i)$ and a free group of rank m. A finite-order element of a free product is conjugate into one of its factors; this group is therefore torsion-free.

Cut (3) along its connected-sum three-spheres. Its universal cover is a tree of universal covers of punctured factors, glued along three-spheres. Removing a locally finite collection of open four-balls does not change H_1 or H_2: excision gives relative groups only in degree four. Consequently each lifted vertex piece has vanishing H_1, H_2. The same holds for the punctured universal cover of $S^1 \times S^3$, whose unpunctured cover is $\mathbb{R} \times S^3$. Mayer–Vietoris over finite subtrees, followed by the direct limit, gives $H_2(\widetilde{M};\mathbb{Z}) = 0$; edge three-spheres have no first or second homology. Compact singular cycles involve only finitely many pieces. Hurewicz now gives $\pi_2(M) = 0$. □

For a concrete distinction, take

$$M = (\Sigma_g \times \Sigma_h) \# m(S^1 \times S^3), \qquad g, h \geqslant 2, \quad m > 0. \tag{4}$$

It satisfies (H), but is not aspherical. If it were aspherical it would be a $K(\pi, 1)$ for $\pi = \pi_1(\Sigma_g \times \Sigma_h) * F_m$. Another finite $K(\pi, 1)$ is $(\Sigma_g \times \Sigma_h) \vee \bigvee_{i=1}^{m} S^1$, of Euler characteristic $\chi(\Sigma_g \times \Sigma_h) - m$. In contrast the connected sum (4) has Euler characteristic $\chi(\Sigma_g \times \Sigma_h) - 2m$. Homotopy invariance contradicts $m > 0$. Thus Theorem 1 is a genuine enlargement of the topological scope, not another definition of asphericity.

3. Consequences for compactness and exact realization

Corollary 3. Every assertion in compact-regime realization, including its critical equiintegrability version, remains valid with "aspherical" replaced by (H), with the same stated geometric dependencies.

Proof. In that proof the geometric inputs are Ricci bounds, diameter, volume, and one-sided Weyl control. Harmonic-radius blow-up produces a smooth nonflat half-flat Ricci-flat ALE space and embeds its compact core in the original manifold. Asphericity enters only in forbidding that embedding. Apply Theorem 1 instead. The later diffeomorphism compactness and Einstein-limit argument are unchanged. □

In particular, for $q > 2$, fixed $D, v, \Lambda > 0$, and topology (H),

$$\operatorname{diam}(g) \leqslant D, \quad V_g \geqslant v, \quad \|W_g^+\|_{L^q} \leqslant \Lambda, \quad \|\operatorname{Ric}_g^\sharp + I\|_{\infty,\mathrm{op}} < \varepsilon(D, v, \Lambda, q) \implies \exists h : \operatorname{Ric}_h = -h. \tag{5}$$

The conclusion concerns the same smooth manifold. It does not assume that the input is close to a previously known Einstein metric.

The theorem also supplies the topological step in the new critical chiral threshold and gauge-saturation realization. Those notes prove their analytical inputs independently of Proposition 2.

4. An application to handle additions with actual gauge supply

For (4), let $A = 8(g-1)(h-1) > 0$. The canonical SW class on the product, followed by the handle surgery formula retaining the degree-one insertions, supplies an actual conformally solvable class with

$$a^2 = A, \qquad (a_g^+)^2 \geqslant A. \tag{6}$$

The supply statement, not merely this intersection calculation, is the classical input detailed in the polarized closure, Sections 1–2. Here $b_+ > 1$, so no chamber choice is necessary.

If m is odd, then $b_1(M) = 2g + 2h + m$ is odd. No complex structure on this smooth manifold can be Kähler, in either orientation, since a compact Kähler manifold has even first Betti number. Apply the gauge-saturation realization theorem with its extension (H). For every fixed diameter bound D, it follows that

$$\delta_-(g) + \frac{\mathfrak{D}_2(g, a)}{V_g} \geqslant \varepsilon(D, A_0) > 0, \qquad (a_g^+)^2 \geqslant A_0 > 0, \quad \operatorname{diam}(g) \leqslant D. \tag{7}$$

One may choose $A_0 = 8$ simultaneously for all products in (4). When the sum on the left is small, the scalar monopole bound supplies the volume lower bound used by the realization theorem; it is not an additional hypothesis on these examples.

For instance $g = 2, h = 11, m = 1$ gives $A = 80$, $(2\chi + 3\tau)(M) = 76$, and handle ratio $4m/A = 1/20$. The old displayed one-third-band inequality with constant at most $1/4$ is vacuous at zero defect for this ratio, since $1/3 - 1/20 = 17/60 > 1/4$. Statement (7) is a different conclusion: it forbids joint gauge saturation and pointwise Einstein approximation under a diameter bound. It does not replace that old unrestricted integral gap or prove a gap for δ_- alone.

No assertion is made that every topology in this infinite family admits metrics satisfying a common diameter bound and very small Ricci defect. Indeed (6) and volume comparison restrict the available topologies when D is fixed. Nor does (7) exclude all non-Kähler Einstein metrics: such a metric could have positive $\mathfrak{D}_2/V$. Its precise increment is an application of a new exact-realization implication to nonaspherical smooth topology.

5. Dependencies and further scope

Theorem 1 uses Wright, *Quotients of gravitational instantons* and Suvaina, *ALE Ricci-flat Kähler metrics and deformations of quotient surface singularities*. The actual gauge supplier in Section 4 is the one in LeBrun, Proposition 3.2, with the handle insertions recorded in the inherited closure proof.

The condition (H) is sufficient, not proved necessary. More generally the same argument works whenever finite core groups map trivially into $\pi_1(M)$ and the lifted ambient has the two homology vanishings used above. Manifolds with nonzero π_2, torsion in π_1, or unbounded-diameter collapse are not settled by this extension.